quick sort.

(3) Mathematical computations : Recursive definitions naturally model many mathematical problems.

Ex:- • factorial calculation
• fibonacci series
• Greatest common Divisor

(4) Backtracking problems : It helps to explore all possible solutions are backtrack when needed

Ex:- • sudoko solver
• Maze solving
• knights tour

(5) Divide & conquer Algorithms : large problems are divided into smaller subproblems that are solved recursively

Ex: Merge sort
Quick sort

(6) Graph Algorithms : used in graph traversal techniques

Ex: Depth-first search
Topological sorting
finding connected components

(7) Dynamic programming : recursive solutions are optimized using memoization

Ex: fibonacci with memoization
longest common subsequences (LCS)

(8) file system Operators : Directories often contain subdirectories, making recursion a natural choice.

Ex:- listing all files in a directory tree
Searching for a file in nested folders

(9) parsing and compilers : Nested languages structures

Ex:- expression evaluation
syntax tree construction
Recursive descent parsing